TEAMFARE

AI-assisted planning and coordination for team outings

Research venues, agree on a time, approve a budget and keep the expense record together.

Challenge: Powering the Future of Work
Participation: Online · November 2026

teamfare.dixmail.xyz

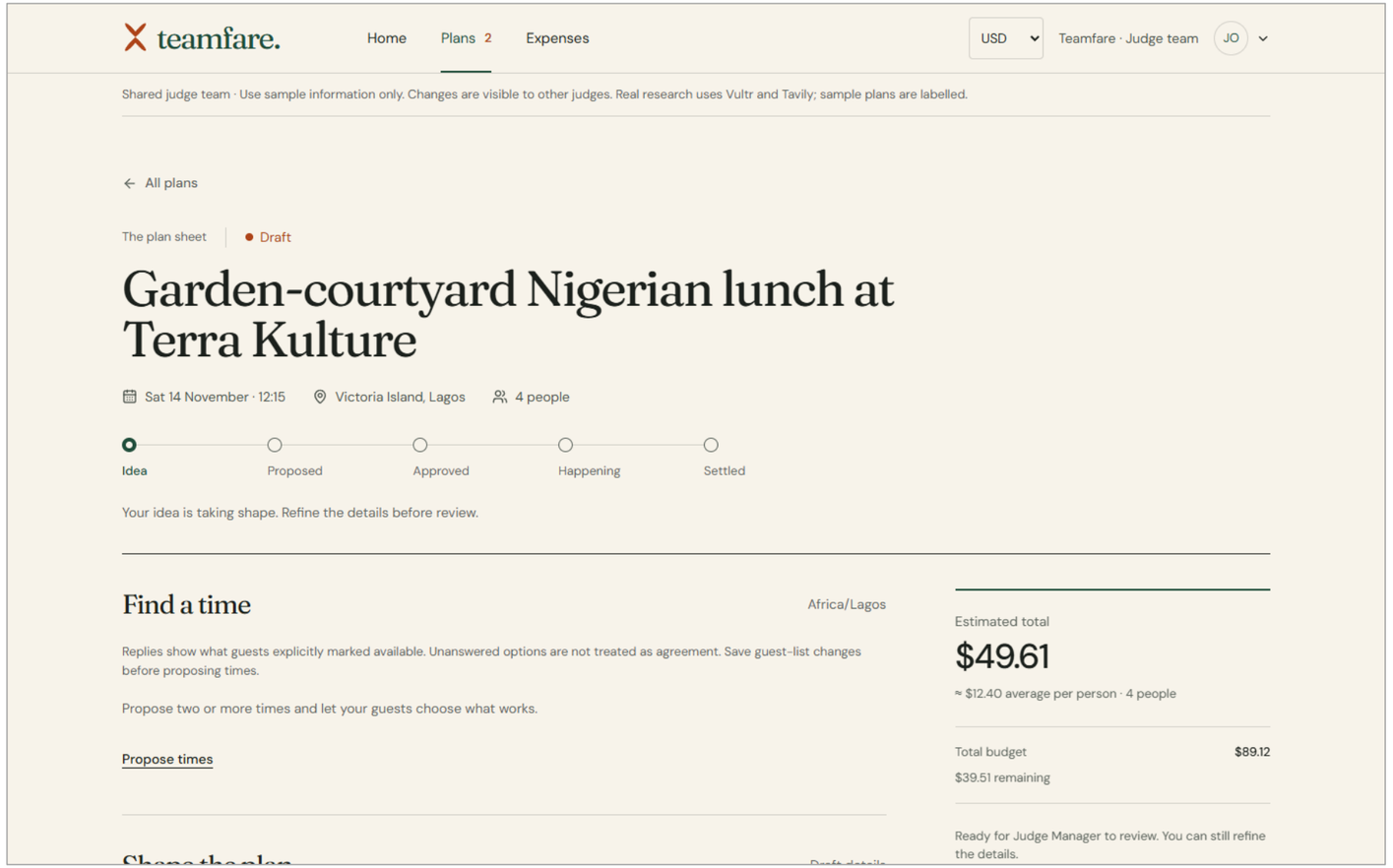


Figure 1. A real Plan Sheet saved from sourced research.

PROBLEM STATEMENT AND PROJECT OBJECTIVE

A team outing combines employee engagement with operational work. Decisions often move between messages, calendars and spreadsheets.

1. Planning information is scattered

Organisers compare venues, collect availability and follow up on attendance through separate conversations.

2. Spending needs a clear decision

Managers need the estimated total, allowance and approval history before a team proceeds.

3. Expense evidence arrives afterwards

Receipts and spending entries must be collected and consolidated into a report.

PROJECT OBJECTIVE

Provide a shared workflow that carries an outing from researched options through scheduling, manager review, attendance and expense reporting.

SOLUTION OVERVIEW: THE SHARED PLAN SHEET

Teamfare stores the outing in one document. The organiser, manager and guests each act on the same current plan.

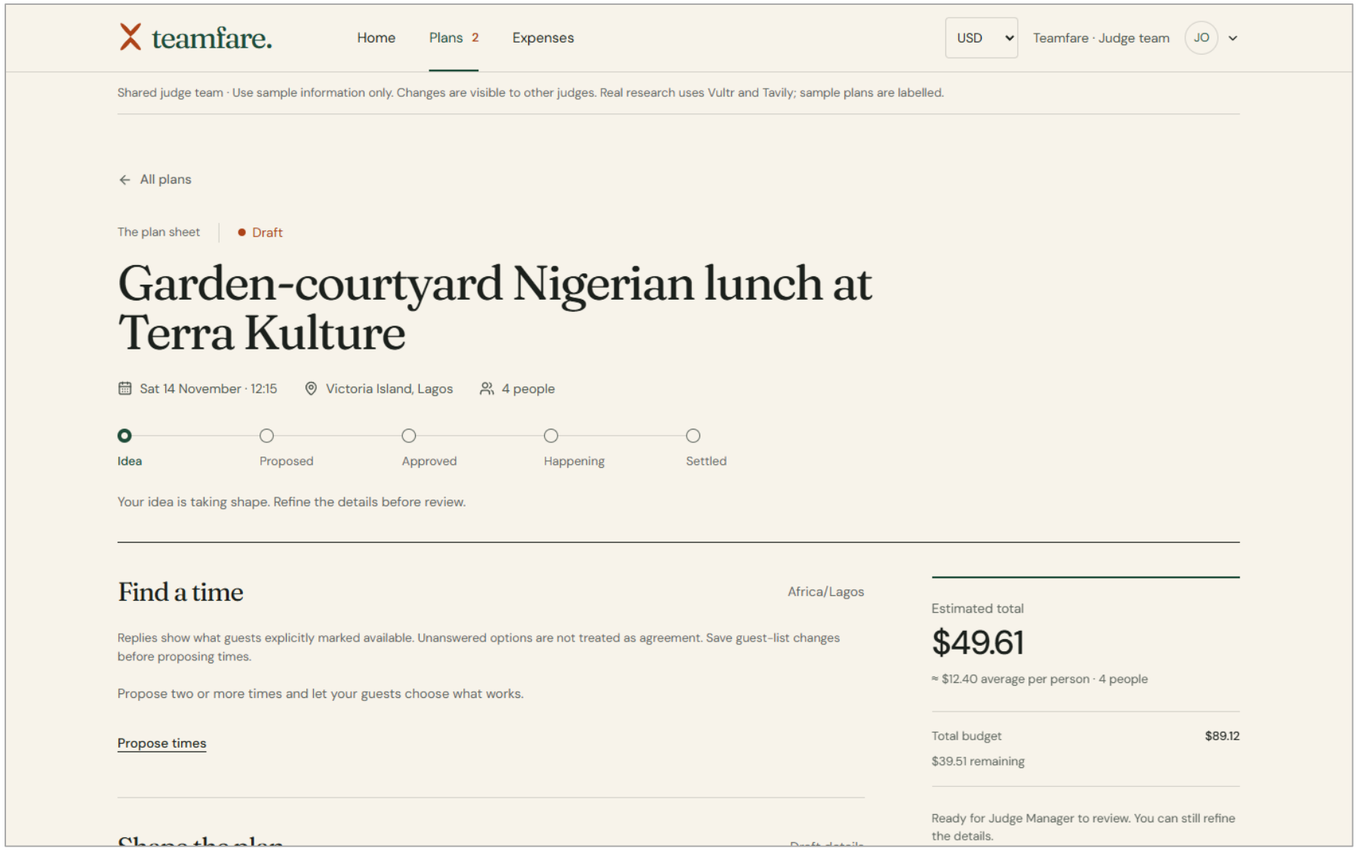


Figure 2. Live researched draft with named guests and a cost summary.

A complete proposal

Date, place, headcount, itinerary, cost assumptions and allowance stay together.

Role-based actions

Organisers refine; a separate manager reviews; named guests reply and RSVP.

A retained record

Status changes, review notes and recorded expenses remain attached to the outing.

Idea → **Proposed** → **Approved** → **Happening** → **Settled**

AI RESEARCH: QWEN, VULTR AND TAVILY

The agent turns the brief into researched options. The organiser chooses which option becomes an editable plan.

Reasoning and tool calls

Qwen runs through Vultr Serverless Inference and requests bounded research actions.

Public-source research

Tavily searches and extracts pages. Options retain citations, estimates and questions to confirm.

Deterministic totals

Server code calculates costs. Fixed and per-person lines remain editable before review.

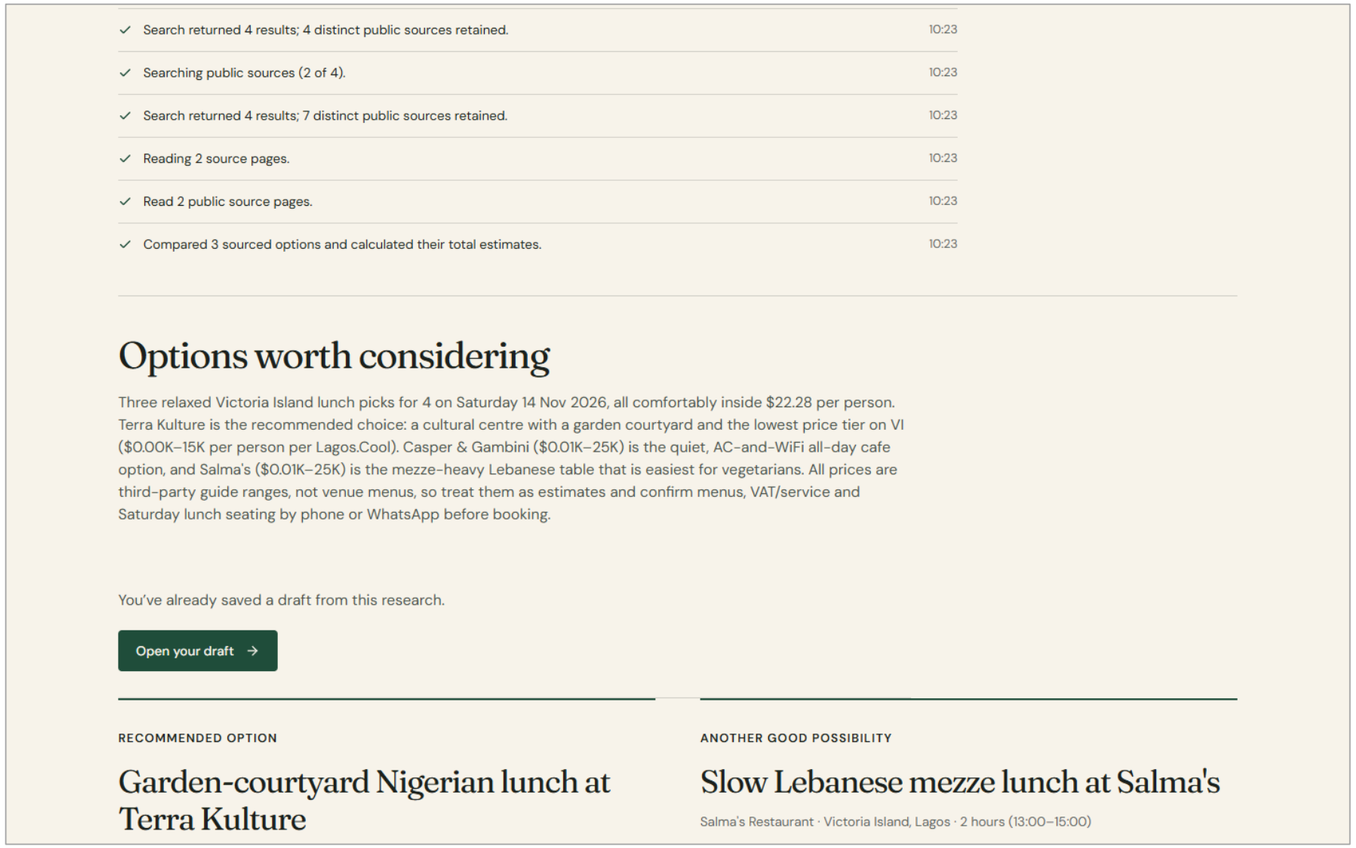


Figure 3. Actual venue options from the deployed Qwen/Vultr/Tavily run.

Observed run, 5 Nov 2026: 2 searches · 2 pages read · 3 sourced options.

SCHEDULING AND TEAM COORDINATION

Availability is based on explicit teammate replies. Unanswered invitations remain visible instead of being treated as agreement.

Find interested teammates

Saved activity interests help organisers select the guest list.

Compare proposed times

Guests mark which slots work. Ranking separates available, unavailable and unanswered guests.

Coordinate after approval

Opted-in invitations and reminders support RSVP and calendar downloads.

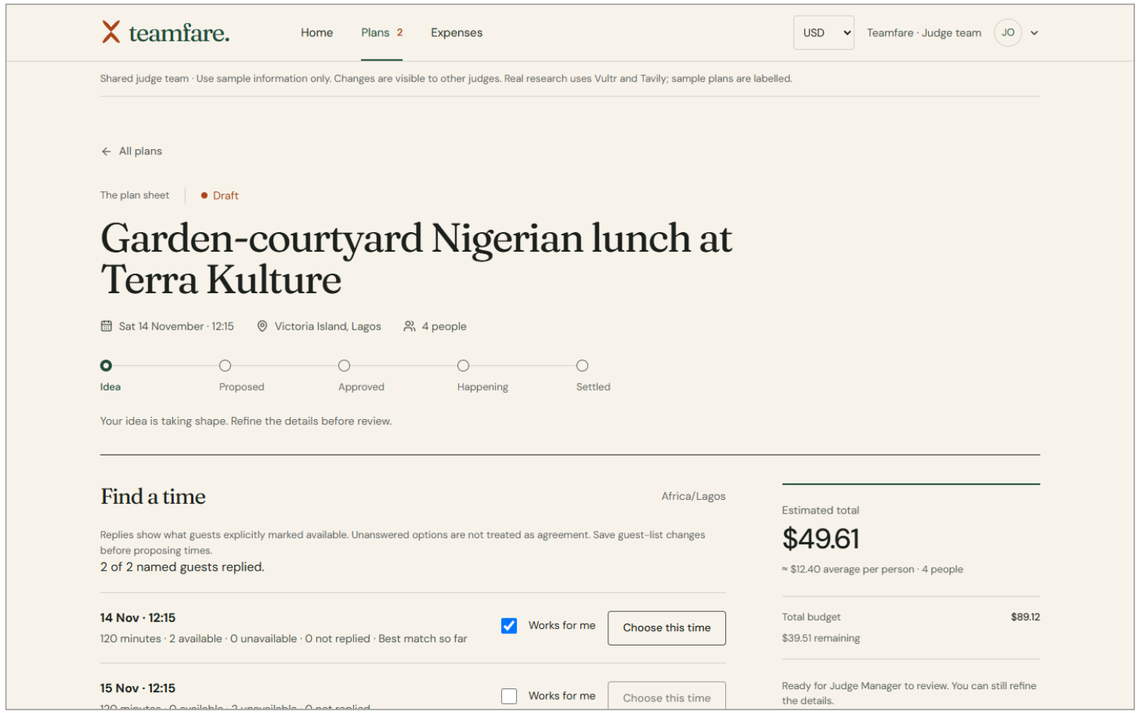


Figure 4. Two named demonstration guests have replied to the proposed times.

The organiser chooses a time and acknowledges any conflicts or missing replies. Calendar files use the saved date, start time and duration.

MANAGER REVIEW AND BUDGET CONTROL

A separate manager reviews the proposal before approval. The review exposes timing, headcount, estimated cost and the remaining allowance.

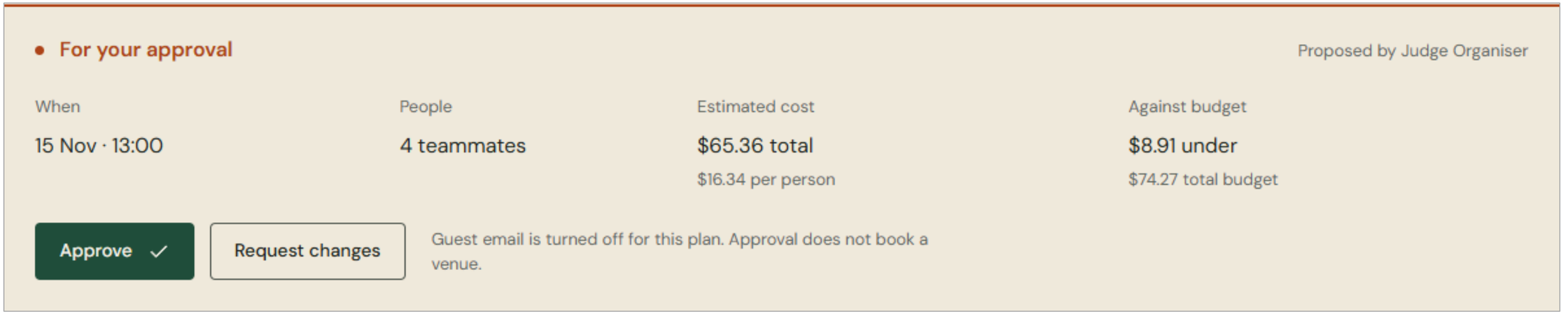


Figure 5. Live review panel for a sample four-person proposal.

Independent approval

The organiser cannot approve their own proposal. A manager records the decision.

Actionable feedback

Request changes with a required note. The organiser revises and resubmits.

Server-side checks

Over-budget approval and stale revisions are rejected. Activity records the actual actor.

EXPENSE RECORDING AND REPORTING

Recorded spending and its supporting files remain attached to the approved outing, ready for the manager’s report.

Private receipt evidence

PDF, PNG and JPEG uploads have size limits. Access is restricted to the expense creator, payer and managers.

Consistent expense reporting

Filter records by outing and inclusive dates. Totals and formula-safe CSV use the same selected rows.

Explicit report sharing

A private-team manager can email the filtered report to a verified teammate. Real delivery was tested separately.

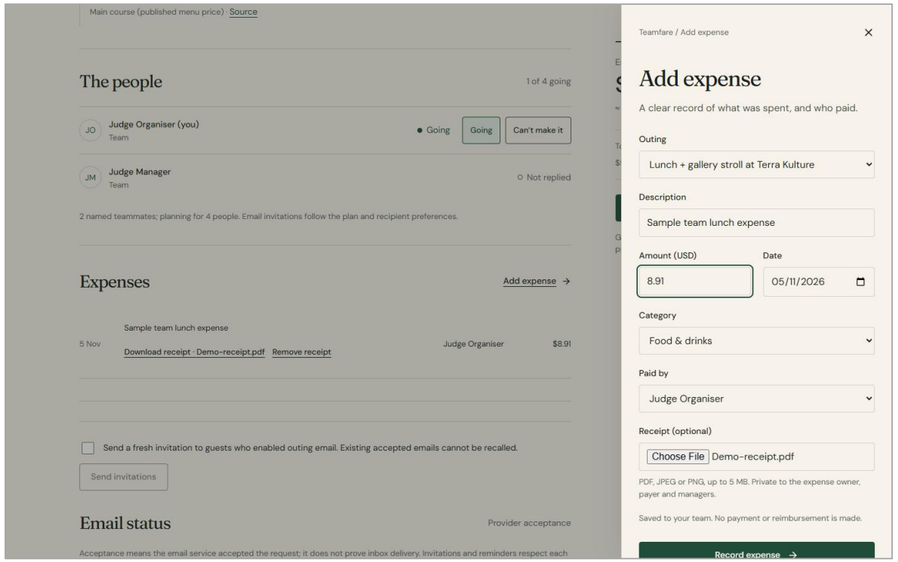
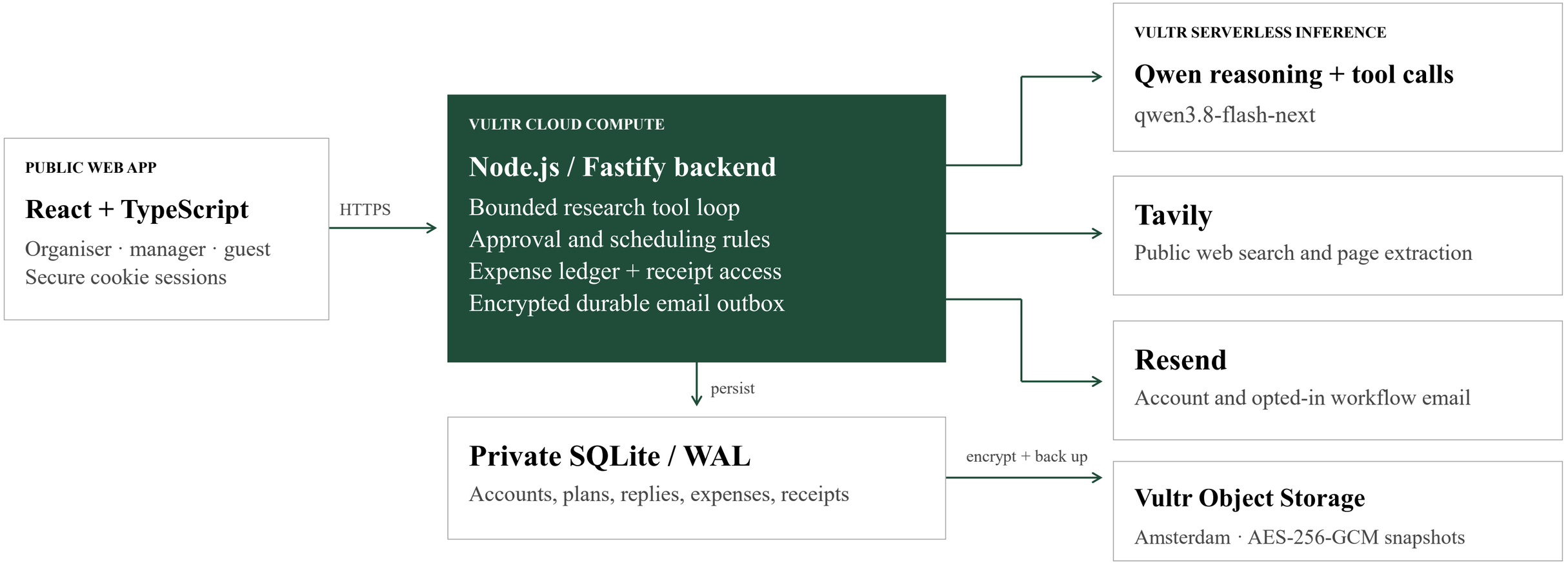


Figure 6. Receipt upload form (sample entry, not submitted).

SYSTEM ARCHITECTURE

Vultr is the operating core: Cloud Compute hosts the application and workflow engine, Serverless Inference serves Qwen, and Object Storage protects recovery.



Docker + Caddy on one Vultr VM · HTTPS · private persistent data · independent uptime checks · operator-led recovery

SECURITY AND OPERATIONAL RELIABILITY

The model researches options; authenticated product rules control the decisions, records and external messages.

Account and team protection

Normal accounts verify email ownership and use password-plus-code sign-in. Origin, CSRF and team-scope checks protect requests.

Bounded agent work

Search, source reading and option submission have time, call and usage limits. Source IDs and price claims are validated.

Workflow integrity

Revision checks reject stale writes. Transactions and encrypted durable email jobs support retries and duplicate protection.

Backup and recovery

Encrypted off-VM backups include receipt bytes. Downloads, decryption, integrity and restore checks have been exercised.

Public judge accounts are deliberately isolated and restricted; ordinary accounts retain email verification.

IMPLEMENTATION AND VALIDATION RESULTS

Verification combines automated checks with real deployed accounts, providers, restart persistence and offsite recovery.

AREA	RESULT	WHAT WAS EXERCISED
API and security	**31 checks passed**	Authentication, permissions, revisions, receipts, outbox retries, calendar/CSV serialization and backups.
Browser workflows	**17 journeys passed**	12 preview journeys, 4 account/workflow journeys and 1 fixture-backed research journey.
Real integration	**4 verified users**	Actual email proofs, research, scheduling, review, RSVP, private receipts and reporting.
Read workload	**160 reads; 0 failures**	16 requests in flight across 4 users. Median 176 ms; p95 527 ms in one measured run.
Restart persistence	**Restart verified**	Sessions, RSVP, receipt bytes and 8 reminders persisted. Cancellation stopped all 8.
Offsite recovery	**Independent restore**	Encrypted objects restored on the developer’s PC; receipt hashes and original accounts matched.

Scope: controlled technical checks and a small-team read test; no claim of zero bugs or general capacity.

CONCLUSION AND PROJECT ACCESS

Teamfare delivers a working research → scheduling → approval → attendance → expense-reporting workflow. Employees organise time together while managers retain a clear spending record.

Vultr runs the web application, backend, model inference and encrypted backup storage. Venue bookings and payments remain external; calendar files are exports.

Live app: https://teamfare.dixmail.xyz

Repository: https://github.com/ELLA0VICTOR/teamfare

PUBLIC JUDGE ACCESS — use separate browser profiles for the two roles

ROLE	EMAIL	PASSWORD
Organiser	organiser@judges.teamfare.invalid	TeamfareOrganiser2026!
Manager	manager@judges.teamfare.invalid	TeamfareManager2026!

Shared demonstration team: use sample information. Normal accounts require email verification and sign-in codes.